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Departamento de Engenharia Informática

Segurança em Tecnologias da Informação, **MEI 2020/2021**

Exame – Época Normal (1h45)

14 Jun 2021

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1. Consider the email security solutions currently employed on the Internet. If a certain domain is using DKIM (DomainKeys Identified Mail), does this make the use of S/MIME by the users of the same domain unnecessary? Justify your answer.

2. Consider the DES (Data Encryption Standard) symmetric encryption algorithm. What is the relationship between the avalanche effect exhibited by this algorithm and the principles defined in Feistel's structure to make cryptanalysis attacks more difficult? Justify your answer.

3. Consider the CBC (Cipher Block Chaining) mode of block encryption. In this mode, is it advisable to reuse a certain IV (Initialization Vector) value with the same encryption key? Justify your answer.

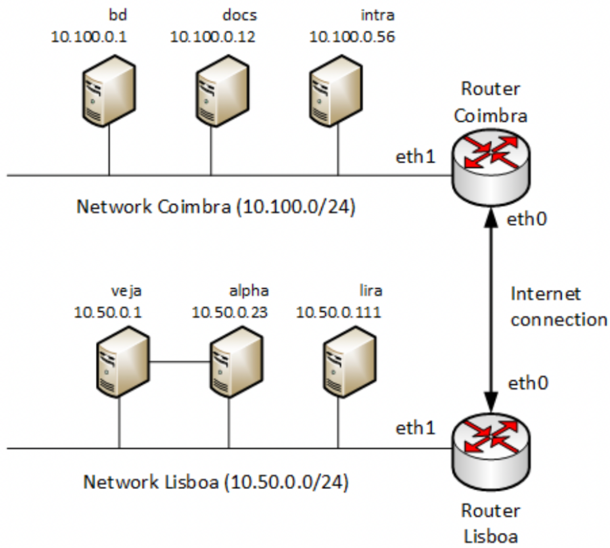
4. Consider the RSA algorithm. Demonstrate the operation of this algorithm, considering the usage of the two prime numbers p=7 and q=13 and the encryption of a message represented by the number "2".

5. Consider the IPSec protocol, the ESP security header, and the tunnel operation mode. Describe an application scenario where two VPN tunnels with encryption (both tunnels using the ESP header) are invalidly combined. Justify your answer.

6. Why MAC (Message Authentication Code) codes cannot guarantee non-repudiation? Justify your answer.

7. Consider the Kerberos Protocol. Why, in this protocol, the system clock of the various participating hosts must be synchronized or, at best, with a small deviation between them? Justify your answer.

8. Consider the network scenario illustrated in the following Figure. In this scenario, two networks (Coimbra and Lisboa) are interconnected via the. Consider also that both routers, as well as all the various hosts, are using the Linux operating system.



Note: in the following exercises, if necessary, you should assign any missing IP addresses.

a) Consider that your goal is to use IPTables to setup a filtering policy on *host* "bd". Indicate the IPTables commands required to activate the following filtering policy in this host, for communications entering the system (using the INPUT chain):

- Authorize connections to the SSH service (TCP/Port 22) originating from the Lisbon network.
- Authorize connections to the PostgreSQL service (TCP/Port 5432) originating from the "intra" machine.
- Authorize the communications necessary for the server to operate as client of the DNS and NTP services (available on other hosts).
- Reject all other IP communications entering the system, which are not necessary for the correct functioning of the previous services.

b) Consider now that your goal is to use IPTables to define a filtering and NAT policy on the Lisbon router. Also consider that, on this router, the policy of the FORWARD chain is already set to "DROP". Indicate the IPTables commands required to activate the following NAT policy:

- Implement SNAT for all communications originating in the Lisbon Network and destined for the SSH and HTTP services of the host "bd".
- The SSH connections made to the *eth0* interface IP address of the Lisbon router must be redirected to the same service, but on host "lira".